

■ C Program: Student Marksheet with Grade Calculation

```
#include <stdio.h>

int main() {
    char name[50];
    int regno;
    int marks[5];
    float total = 0, average;
    char grade;

    printf("=====\n");
    printf("          STUDENT MARKSHEET          \n");
    printf("=====\n");

    printf("Enter Student Name: ");
    scanf("%s", name);
    printf("Enter Register Number: ");
    scanf("%d", &regno);

    printf("\nEnter marks for 5 subjects (0 to 100):\n");
    for (int i = 0; i < 5; i++) {
        printf("Subject %d: ", i + 1);
        scanf("%d", &marks[i]);

        if (marks[i] < 0 || marks[i] > 100) {
            printf("Invalid input! Marks should be between 0 and 100.\n");
            return 1;
        }
        total += marks[i];
    }

    average = total / 5;

    if (average >= 90)
        grade = 'A';
    else if (average >= 80)
        grade = 'B';
    else if (average >= 70)
        grade = 'C';
    else if (average >= 60)
        grade = 'D';
    else if (average >= 50)
        grade = 'E';
    else
        grade = 'F';

    printf("\n=====\n");
    printf("          MARKSHEET RESULT          \n");
    printf("=====\n");
    printf("Name           : %s\n", name);
    printf("Register Number : %d\n", regno);
    printf("-----\n");
    for (int i = 0; i < 5; i++)
        printf("Subject %d Marks : %d\n", i + 1, marks[i]);
    printf("-----\n");
    printf("Total Marks      : %.2f\n", total);
    printf("Average Marks    : %.2f\n", average);
    printf("Grade Awarded    : %c\n", grade);
    printf("=====\n");

    if (grade == 'A')
        printf("Excellent performance! Keep it up!\n");
    else if (grade == 'B' || grade == 'C')
        printf("Good job! You can do even better!\n");
    else if (grade == 'D' || grade == 'E')
        printf("Needs improvement. Work harder!\n");
    else
        printf("Fail. Please study well for next time.\n");

    printf("=====\n");
    return 0;
}
```

